Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/27/2016
PI Name:	Diddams, Scott A. Ph.D.		
Project Title:	Development of an Optical Microcomb System in Support of Fundamental Physics Research on the International Space Station		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/01/2012	**End Date:**	05/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/31/2018 per PI (Ed., 3/7/17)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Papp, Scott (National Institute of Standards & Technology) Vahala, Kerry (California Institute of Technology)		
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Task Description:	Our research program will investigate and develop advanced microresonator optical frequency comb technology in support of planned European Space Agency (ESA) activities on the International Space Station (ISS). Optical frequency combs, which provide a dense set of absolute frequency reference lines, are an integral component of modern optical atomic clocks. They provide a link between optical and microwave signals and a means to access the stability of optical clocks at any wavelength. For example, a frequency comb will be required for the Space Optical Clock (SOC) experiment to achieve all of its fundamental scientific goals by providing the critical phase coherent optical or microwave link between the atomic frequency standard on the ISS and ground-based clocks. The same frequency comb technology will be a required component of other proposed ESA and NASA fundamental-physics clock experiments, as well as an enabling technology for ESA's planned Quantum Weak Equivalence Principle (QWEP) atom interferometry experiments. Microresonator combs (microcombs) can be lithographically patterned and fabricated on a single silicon chip with dimensions of a few square centimeters and therefore offer revolutionary advantages beyond any other conventional laser comb technology in terms of size, power consumption, and sensitivity to their environment. However, the full capability of a microcomb for precision optical frequency metrology has not been realized. The laboratory experiments we propose will, for the first time, characterize a microcomb's stability with respect to known optical standards, thus verifying its suitability to support the high-performance optical clocks envisioned for SOC and other space missions. Our microcomb system will be tested and interfaced with state-of-the-art optical clocks based on strontium and ytterbium atoms, which are the two candidate SOC platforms. Furthermore, the microcomb will have an output at 1064 nm and a countable microwave repetition rate in the 10-50 GHz range, which are compatible with the optical and microwave links required to achieve the fundamental physics goals of the SOC mission. Our research will develop advanced technology to enable full on-chip integration of a microcomb with essential components for time and frequency metrology including, high-Q silica microresonators, clock-signal coupling, comb self-referencing, and fiber v-groove end-couplers. These efforts will contribute to the long-term goal of realizing a single chip-scale device operating at the convergence of optics, frequency metrology, opto-mechanics, and microwave technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The best optical clocks now keep time with uncertainty measured in the 18th digit; however, their complexity constrains such remarkable precision to a research laboratory setting. Our research is focused on developing key components of optical clocks that will eventually enable their use outside the lab and in outer space. In particular, we aim to develop the compact “clockwork” that will enable the counting of optical cycles on the femtosecond time scale, as well as the synthesis of optical and microwave frequencies critical for transmission and comparison of multiple atomic clocks. Precision time keeping, as explored in our research, is anticipated to impact a wide variety of human activities both in Earth and space. Not only do clocks provide an accurate tool to analyze the inner workings of atoms and molecules, but they are critical to communication and navigation systems that we rely on every day. Examples include the global positioning system and cellular phone networks. Our research is an important step in developing the next generation clocks which will not only be more accurate, but also more compact and easily fabricated.
Task Progress:	Our National Institute of Standards & Technology (NIST)/Caltech research program is focused on developing a revolutionary optical frequency comb system-on-a-chip in support of future experiments that leverage optical frequencies on the International Space Station (ISS) as well as other space-based platforms. Optical frequency combs are a critical component of proposed space missions on the ISS such as Space Optical Clock (SOC) and Quantum Weak Equivalence Principle (QWEP). The system we are developing, based on parametric comb generation in microresonators (a “microcomb”), offers unique advantages, including small size and power consumption, monolithic construction to reduce environmental sensitivity, and the potential for lithographic integration on silicon. Optical frequency combs, which provide a dense set of absolute frequency reference lines, are an integral component of modern optical atomic clocks. They provide a link between optical and microwave signals and a means to access the stability of optical clocks at any wavelength. For example, a frequency comb will be required for the Space Optical Clock (SOC) experiment to achieve all of its fundamental scientific goals by providing the critical phase coherent optical or microwave link between the atomic frequency standard on the ISS and ground-based clocks. The same frequency comb technology will be a required component of other proposed ESA and NASA fundamental-physics clock experiments, as well as an enabling technology for ESA's planned Quantum Weak Equivalence Principle (QWEP) atom interferometry experiments. Microresonator combs (microcombs) can be lithographically patterned and fabricated on a single silicon chip with dimensions of a few square centimeters and therefore offer revolutionary advantages beyond any other conventional laser comb technology in terms of size, power consumption, and sensitivity to their environment. However, the full capability of a microcomb for precision optical frequency metrology has not been realized. The laboratory experiments we are undertaking will characterize a microcomb's stability with respect to known optical standards, thus verifying its suitability to support the high-performance optical clocks envisioned for SOC and other space missions. Additionally, we are developing the tools and techniques required to stabilize microcomb systems relative to state-of-the-art optical clocks. Critical to this stabilization is the self-referencing of a microcomb, which is the key step to enable optical-to-microwave synthesis with a frequency combs. These efforts will contribute to the long-term goal of realizing a single chip-scale device that can perform broad bandwidth frequency synthesis for optical timekeeping, precision metrology, and fundamental physics investigations. In the third and fourth years of the program, our research activities were primarily focused on the self-referencing of a silica chip microcomb. This turned out to be a significant undertaking that includes (1) the generation of phase-locked microcomb spectra, (2) coherent spectral broadening to an octave, (3) self-referencing via nonlinear interferometry with low microcomb powers, and (4) implementation of the servo control of the microcomb. Significantly, our effort over the past two years resulted in the first demonstration of the use of a microcomb to provide a phase-coherent link between optical and microwave domains, thereby validating its low-noise synthesis capabilities.
Bibliography Type:	Description: (Last Updated: 05/13/2021)

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